

1. (a) The likelihood function in this case is

$$L(\alpha) = \prod_{i=1}^n f_{X_i}(x_i; \alpha) = \frac{\alpha^n \lambda^{n\alpha}}{(\prod_i x_i)^{\alpha+1}}, \quad x_i > \lambda$$

and hence the log likelihood is

$$\ell(\alpha) = n \log \alpha + n\alpha \log \lambda - (\alpha + 1) \sum_i \log(x_i).$$

We find that

$$\frac{\partial \ell(\alpha)}{\partial \alpha} = \frac{n}{\alpha} + n \log \lambda - \sum_i \log(x_i).$$

Setting this equal to zero, and solving, we find that

$$\hat{\alpha} = \frac{n}{\sum_{i=1}^n [\log(X_i) - \log \lambda]} = \frac{n}{\sum_{i=1}^n \log(X_i) - n \log \lambda}.$$

The second derivative of the log likelihood is $-n/\alpha^2$, which is negative for all α , so we have indeed found a maximum, and $\hat{\alpha}$ is the MLE.

- (b) We have that under certain regularity conditions, the distribution of $\hat{\alpha}$ is approximately $N(\alpha_0, 1/nI(\alpha_0))$. So we must calculate the Fisher information $I(\alpha)$.

$$\begin{aligned} f_X(x; \alpha) &= \frac{\alpha \lambda^\alpha}{x^{\alpha+1}} \\ \log f_X(x; \alpha) &= \log \alpha + \alpha \log \lambda - (\alpha + 1) \log x \\ \frac{\partial}{\partial \alpha} \log f_X(x; \alpha) &= \frac{1}{\alpha} + \log \lambda - \log x \\ \frac{\partial^2}{\partial \alpha^2} \log f_X(x; \alpha) &= -\frac{1}{\alpha^2} \\ I(\alpha) &= E \left(-\frac{\partial^2}{\partial \alpha^2} \log f_X(X; \alpha) \right) = \frac{1}{\alpha^2} \end{aligned}$$

Hence, the distribution of $\hat{\alpha}$ is approximately $N(\alpha_0, \alpha_0^2/n)$. We would approximate the variance as $\hat{\alpha}^2/n$.

Note: We had a more direct route to this answer. We found in part (a) that the second derivative of the log likelihood is $-n/\alpha^2$. Thus, we know that

$$nI(\alpha) = E \left(-\frac{\partial^2 \ell(\alpha)}{\partial \alpha^2} \right) = n/\alpha^2.$$

- (c) Recall that

$$\hat{\alpha} = \frac{n}{\sum_{i=1}^n [\log(X_i) - \log \lambda]} = \frac{n}{\sum_{i=1}^n \log(X_i) - n \log \lambda}.$$

Plugging in $n = 100$, $\lambda = 2$, and $\sum_{i=1}^n \log(x_i) = 107.39$ yields the value $\hat{\alpha} = 2.63$.

We know that the variance of the MLE is approximately α_0^2/n , from above, so we can use $\hat{\alpha}/\sqrt{n}$ to approximate the standard error of the estimator. In this case, that comes out to be $2.63/\sqrt{100} = 0.263$.

We could now say the following: “Our best estimate of α is 2.63. This was found using an estimator with an approximate standard error of 0.263. Hence, errors in our estimate should be anticipated on the order of 0.263.”

2. (a) The density is (remember that β is a fixed constant)

$$f_X(x; \theta) = x^{\theta-1} \exp(-\beta x) \beta^\theta / \Gamma(\theta)$$

so that

$$\log f_X(x; \theta) = (\theta - 1) \log(x) - \beta x + \theta \log(\beta) - \log(\Gamma(\theta)).$$

the first derivative of this is

$$\log(x) + \log(\beta) - \frac{\Gamma'(\theta)}{\Gamma(\theta)},$$

and second derivative is

$$- \left[\frac{\Gamma(\theta)\Gamma''(\theta) - \Gamma'(\theta)^2}{\Gamma(\theta)^2} \right].$$

One notes that this second derivative is not a function of x , so

$$I(\theta) = \frac{\Gamma(\theta)\Gamma''(\theta) - \Gamma'(\theta)^2}{\Gamma(\theta)^2}.$$

- (b) From above, it follows that the MLE for θ is approximately normal with approximate variance of

$$(nI(\theta_0))^{-1} \approx (nI(\hat{\theta}))^{-1} = \frac{\Gamma(\hat{\theta})^2}{n(\Gamma(\hat{\theta})\Gamma''(\hat{\theta}) - \Gamma'(\hat{\theta})^2)}.$$

- (c) We are considering $g(\theta) = \log(\theta)$, so

$$\frac{\partial g(\theta)}{\partial \theta} = 1/\theta.$$

We know that the MLE for $g(\theta)$ is $g(\hat{\theta})$, and that, from the Δ -method, $g(\hat{\theta})$ is approximately normal with mean $g(\theta)$ and variance

$$g'(\hat{\theta})^2 \left(\frac{\Gamma(\hat{\theta})^2}{n(\Gamma(\hat{\theta})\Gamma''(\hat{\theta}) - \Gamma'(\hat{\theta})^2)} \right) = \left(\frac{\Gamma(\hat{\theta})^2}{n\hat{\theta}^2(\Gamma(\hat{\theta})\Gamma''(\hat{\theta}) - \Gamma'(\hat{\theta})^2)} \right).$$

3. (a) Use the invariance property of the MLE. The MLE for p is $\hat{p} = X/n$, so the MLE for $p/(1-p)$ is $\hat{p}/(1-\hat{p})$.
- (b) First, we need to find the Fisher information $I(p)$. Note that in this case there is **one** iid observation, the single random variable X which has the binomial(n, p) distribution. Hence, in this case

$$f_X(x; p) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, \dots, n$$

and

$$\frac{\partial \log f_X(x; p)}{\partial p} = x/p - (n-x)/(1-p)$$

and

$$\frac{\partial^2 \log f_X(x; p)}{\partial p^2} = -x/p^2 - (n-x)/(1-p)^2.$$

Hence

$$E\left(\frac{\partial^2 \log f_X(X; p)}{\partial p^2}\right) = -np/p^2 - (n-np)/(1-p)^2 = \frac{-n}{p(1-p)}$$

and

$$I(p) = \frac{n}{p(1-p)}.$$

Now, we'll use the Delta method to approximate the distribution of $g(p) = p/(1-p)$. We note that

$$\frac{\partial g(p)}{\partial p} = \frac{1}{(1-p)^2}$$

and thus the Delta method tells us that $g(\hat{p})$ is approximately normal with variance

$$\left(\frac{\partial g(p)}{\partial p} \Big|_{\hat{p}}\right)^2 \frac{1}{n'I(\hat{p})}$$

but keep in mind here that this n' equals one, so this comes out to

$$\left(\frac{1}{(1-\hat{p})^2}\right)^2 \left(\frac{\hat{p}(1-\hat{p})}{n}\right) = \frac{\hat{p}}{n(1-\hat{p})^3}$$

- (c) Using the approximate variance we found above, we can state that a $100(1-\alpha)\%$ confidence interval for $p/(1-p)$ is

$$\frac{\hat{p}}{1-\hat{p}} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}}{n(1-\hat{p})^3}}$$

- (d) Below, we give some Python code for the to test the validity of the confidence interval from part (c).

```
import numpy as np
from scipy.stats import binom, norm

# Define the binomial(n,p) distribution
n = 10
p = 0.10

reps = 10000 # repetitions of the simulation
alpha = 0.05 # value of alpha for the confidence intervals

# Function that returns the CI for the odds

def calcCIodds(x,n,alpha):
    p_hat = x/n
    halfwidth = norm.ppf(1-alpha/2)*np.sqrt(p_hat / (n*(1-p_hat)**3))
    oddsest = p_hat/(1-p_hat)
    return([oddsest-halfwidth,oddsest+halfwidth])

# Simulate the data

x = binom.rvs(n, p,size=reps)

# Calculate the CIs

[lower,upper] = calcCIodds(x,n,alpha)

# What proportion cover the true value of the odds?

np.mean((lower < p/(1-p)) & (upper > p/(1-p)))

0.654
```

In the above simulation, we used the parameter values $n = 10$ and $p = 0.1$, the confidence interval includes the true odds value about 65% of the time. Thus, it is not an adequate confidence interval in this setting. The sample size is too small for the normal approximation to be effective.